BIOCHEMISTRY CUMULATIVE EXAMINATION

February 17, 1990

An enzyme has been isolated which catalyzes a two substrate reaction. You have collected the following data in the presence of two inhibitors X and Y. One of the inhibitors is a structural analog of the variable substrate, A, and the other is not.

substrate (A) mmol/l x 10	velocity umol/min [X]=[Y]=0	velocity umol/min [X]=0.12 mM [Y]=0	velocity umol/min [Y]=0.33 uM [X]=0
5	1.25	0.74	0.48
2.5	0.87	0.45	0.33
1.7	0.67	0.32	0.25
1.2	0.54	0.25	0.2
1	0.45	0.21	0.17
reciprocals of values above			
0.20	0.80	1.35	2.08
0.40	1.15	2.22	3.03
0.59	1.49	3.13	4.00
0.83	1.85	4.00	5.00
1.00	2.22	4.76	5.88

(25) 1. Determine the apparent Michaelis constants K_a and V under the reaction conditions. Why are these apparent constants? How would you determine the true values of K_a and V?

(25) 2. Which inhibitor is the analog of A? How do you know? What is its dissociation constant?

(25) 3. Give one logical explanation for the inhibition of the other inhibitor. Does the inhibition pattern suggest that the reaction is ping pong or sequential? Explain.

(25) 4. Explain how you would determine the dissociation constant of this inhibitor.

The full rate equation for an ordered bi bi reaction is:

$$V = \frac{V_1V_2(AB - PQ/K_{eq})}{K_{ia}K_bV_2 + K_bV_2A + K_aV_2B + V_2AB + K_qV_1P/K_{eq} + K_pV_1P/K_{eq} + V_1PQ/K_{eq} + K_qV_1AP/K_{ia}K_{eq} + K_aV_2BQ/K_{iq} + V_2ABP/K_{ip} + V_1BPQ/K_{ib}K_{eq}}$$

The rate equation for a ping pong reaction is the same except that there is no APB, BPQ or constant term in the denominator.